chiselWare

Dff
Product User Guide

https://github.com/chiselWare/o00-t000-dff

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1 Errata and Known Issues

1.1 Errata

None.

1.2 Known Issues

None.

2 Port Descriptions

The ports for **Dff** are shown below in Table 1. The width of several ports is controlled by the following parameter:

- *width* is the width the d and q ports in bits

Port Name	**Width**	**Direction**	**Description**
clock	1	Input	Positive edge clock
reset	1	Input	Active high reset
enable	1	Input	Enable data
d	*width*	Input	Input to D flip-flop
q	*width*	Output	Output of D flip-flop

Table 1: Port Descriptions

3 Parameter Descriptions

The parameters for **Dff** are shown below in Table 2.

Name	Type	Min	Max	Description
width	Int	1	≥ 1	The data width of the D flip-flop

Table 2: Parameter Descriptions

The Dff is instantiated into a design as follows:

```
// Instantiate small register made of flip-flops
val myRegister = new Dff(width = 8)
```

4 Theory of Operations

4.1 Introduction

The **Dff** is a basic D flip-flop with a parameterized width.

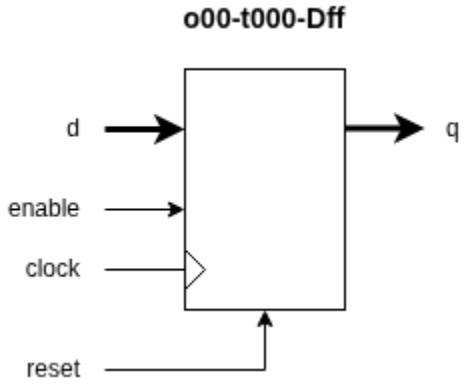


Figure 1: Block Diagram

4.2 Interface Timing

Data on the input **d** will appear on the output **q** on the following clock cycle if **enable** is asserted. If **enable** is not asserted, **q** will retain its current state on the next clock cycle. The timing diagram shown below in Figure 2 represents an instantiation with the following parameters.

```
val myRegister = new Dff(width = 8)
```

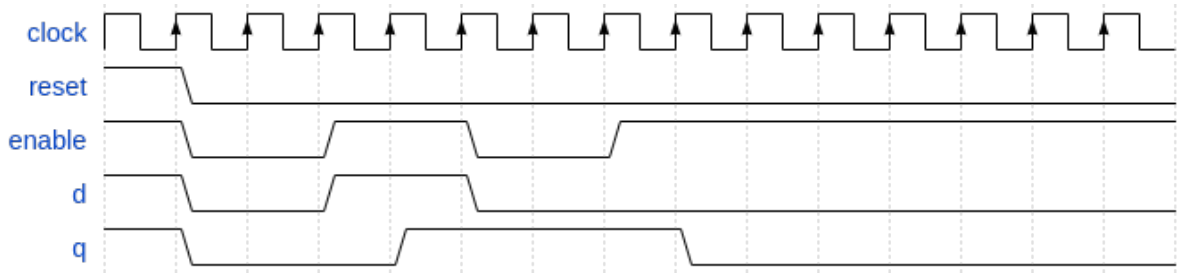


Figure 2: Timing Diagram

5 Simulation

5.1 Tests

The included test bench tests 3 configurations of the Dff. More configurations can be added as desired. For each configuration, random data is applied to the inputs and outputs are checked for correctness.

5.2 Code coverage

The tests exceed the chiselWare minimum standard of 90% Statement coverage and 95% Branch coverage.

5.3 Running simulation

Simulations can be run directly from the command prompt as follows:

```
$ sbt "project core" test
```

or from make as follows:

```
$ make test
```

6 Synthesis

6.1 Area

The **Dff** has been tested in a number of configurations and the following results should be representative of what a user should see in their own technology.

Config Name	width	Gates
small_1	1	8
meduium_64	64	597
large_128	128	1194

Table 3: Synthesis results

6.2 SDC File

An `.sdc` file is generated to provide synthesis and static timing analysis tools guidance for synthesis.

The `Dff.sdc` file can be found in the test case directory as follows:
`./modules/dff/generated/synTestCases/<test case name>/dff.sdc`

6.3 Timing

The following timing was extracted using the generated .sdc files with the open-source Nangate 45nm library.

Config Name	Period	Duty Cycle	Input Delay	Output Delay	Slack
small_1	5ns	50%	20%	20%	3.88 (MET)
medium_64	5ns	50%	20%	20%	3.77 (MET)
large_128	5ns	50%	20%	20%	3.63 (MET)

Table 4: Static Timing Analysis results

6.4 Multicycle Paths

None.